

GROUP THEORY

Set:

A well defined collection of distinct objects is called a set. Sets are usually denoted by capital letters. For example the set of natural numbers is $N = \{1, 2, 3, \dots\}$.

The following table shows the relationship among standard sets.

Complex Numbers (C)				
Real Numbers (R)				Imaginary Numbers
Rational Numbers (Q)			Irrational Numbers (Q')	
Integers (Z)	Vulgar Fractions	Terminating Decimals		
Whole Numbers (W) Natural Numbers (N)				
			Non-terminating Non-recurring Decimals	

Binary Operation:

A binary operation is an operation that applies to two quantities or expressions. For example addition, subtraction multiplication and division.

Groupoid:

A non-empty set which satisfies clousar law with respect to a binary operation is called a groupoid. For example Z with respect to subtraction.

Semi-group:

A groupoid which satisfies associative law is called a semi-group. For example N with respect to addition.

Monoid:

A semi-group which also satisfies identity law is called a monoid. For example W with respect to addition.

Group:

A monoid which also satisfies inverse law is called a group. For example Q with respect to multiplication.

Abelian group:

A group which also satisfies commutative law is called an Abelian group. For example R with respect to multiplication.

The following table shows the information of some standard groups with their binary operations.

Set	Binary Operation
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	Addition
$M_{nm} = \{\text{set of all } n \times m \text{ matrices of same orders}\}$	Matrix addition
$\mathbb{Q} - \{0\}$ $\mathbb{R} - \{0\}$ $\mathbb{C} - \{0\}$ $C_n = \{\text{nth roots of unity}\}$	Multiplication
$S_n = \{\text{Set of all bijective mappings on a set } X \text{ of } n \text{ elements}\}$	Permutation (mapping compositions)
$M_n = \{\text{Set of all } n \times n \text{ non-singular matrix matrices}\}$	Matrix multiplication
$G = \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}$	Matrix multiplication
$\mathbb{Z}_n = \{\text{Set of all residue classes of } n\}$	Modulo addition
$\mathbb{Z}_p = \{\text{Set of all residue classes of } p \text{ where } p \text{ is a prime number}\}$	Modulo multiplication
$P(X) = \{\text{Set of all subsets of } X\}$	Set Intersection
(Klein's four group) $V_4 = \{1, a, b, ab\}$	$a^2 = b^2 = 1$
(Group of quaternions) $H = \{\pm 1, \pm i, \pm j, \pm k\}$ $= \{1, a, a^2, a^3, ab, a^2b, a^3b, b\}$	$a^4 = 1, a^2 = b^2 = (ab)^2$
$D_n = \{\text{Symmetries of } n\text{-polygon}\}$	Mappings compositions
$C_p^\infty = \{\text{Prüfer's } p\text{-group}\}$ $= \{\text{group generated by } a_0, a_1, a_2, \dots\}$	$a_0^p = 1, a_{n+1}^p = a_n$ where p is a prime number and $n = 0, 1, 2, \dots$

Subgroup:

Let G be a group. A non-empty subset H of G is called subgroup of G if H satisfies all the axioms of group with respect to same binary operation of G . For example $(\mathbb{Q}, +)$ is a subgroup of $(\mathbb{R}, +)$.

Cyclic Group:

A group generated by a single element is called cyclic group and that element is called a generator of the group. A group can have more than one generators.

The following table shows the information of some cyclic groups and their generators.

Cyclic group	Generator
$\mathbb{Z}$	1 or -1
$C_2 = \{1, -1\}$	-1

$C_3 = \{1, w, w^2\}$	w
$(C_4 = \{\pm 1, \pm i\})$	i
$C_n =$ group of all n th roots of unity	$e^{2k\pi i/n}$ $K = 1, 2, \dots, n-1.$
$G = \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}$	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
$(\mathbb{Z}_n, +)$	1
$(\mathbb{Z}_n, n)$	$1, \text{ or } -1$
$(\mathbb{Z}_p, Xp), p$ is prime	1

Some Facts About Cyclic Groups

1. Every subgroup of a cyclic group is itself cyclic.
2. A homomorphic image of a cyclic group is itself cyclic.
3. Every cyclic group is Abelian
4. Any two cyclic groups of the same order are isomorphic
5. Let G be a cyclic group of order n . Then G contains one and only one subgroup of order d iff d divides n .
6. Let G be a cyclic group order n and a be generator of G . Then an element $a^k \in G$ is also generator of G iff k and n are relatively prime.
7. Let G be infinite group generated by a . Then a^{-1} is the only other generator of G .
8. Let G be a finite Abelian group of order n . Then G is cyclic iff the exponent of G is n .
9. Every group of prime order is necessarily cyclic.
10. A group G is abelian if and only if the factor group $\frac{G}{Z(G)}$ is cyclic.

Normalizers:

Let X be a non-empty subset of a group G . The set of those elements of G which permute with X is called the normalizer of X in G and is denoted by $N_G(X)$. That is

$$N_G(X) = \{a \in G: aX = Xa\}$$

Centralizer:

Let X be a non-empty subset of a group G . Then set of those elements of G which are permutable with every element of X is called centralizer of X in G . It is denoted by $C_G(X)$. That is

$$C_G(X) = \{a \in G: ax = xa \forall x \in X\}.$$

The following table shows the information of normalizers and centralizers of subsets of some groups.

Group	Subset	Normalizer	Centralizer
Cube roots of unity $C_3 = \{1, W, W^2\}$	$\{1, W\}$	C_3	C_3
	$\{W, W^2\}$	C_3	C_3
Klein's four group $V_4 = \{1, a, b, ab\}$	$\{1, a\}$	V_4	V_4
	$\{1, b, b\}$	V_4	V_4
Symmetric group $S_3 = \{1, a, a^2, b, ab, a^2b\}$	$\{1, b\}$	$\{1, b\}$	$\{1, b\}$
	$\{1, a, a^2, a^3\}$	S_3	$\{1, a, a^2\}$
Dihedral group $D_4 = \{1, a, a^2, a^3, b, ab, a^2b, a^3b\}$	$\{1, a, a^2, a^3\}$	D_4	$\{1, a, a^2, a^3\}$
	$\{1, a^2\}$	D_4	D_4
Group of order 12 $G = \langle a, b, c: a^3 = b^2 = (ab)^2 = c^2 = (bc)^2 = 1, ac = ca \rangle$	$\{1, a, a^2, b, ab, a^2b\}$	G	$\{1, c\}$

Centre of a group:

The centre of a group G is the set of those elements of G which commute with every element of G . That is $Z(G) = \{a \in G: ag = ga \forall g \in G\}$.

The following table shows the information of centers of some standard groups.

Group	Centre
$(\mathbb{Z}, +)$	$\mathbb{Z}$
$(\mathbb{Q}, +)$	$\mathbb{Q}$
$(\mathbb{R}, +)$	$\mathbb{R}$
$(\mathbb{C}, +)$	$\mathbb{C}$
$(\mathbb{Q} - \{0\}, \times)$	$\mathbb{Q} - \{0\}$
$(\mathbb{R} - \{0\}, \times)$	$\mathbb{R} - \{0\}$
$(\mathbb{C} - \{0\}, \times)$	$\mathbb{C} - \{0\}$
$C_3 = \{1, w, w^2\}$	C_3
$V_4 = \{1, a, b, ab\}$	V_4
$S_3 = \{1, a, a^2, b, ab, a^2b\}$	$\{1\}$
$D_4 = \{1, a, a^2, a^3, b, ab, a^2b, a^3b\}$	$\{1, a^2\}$
$H = \{\pm 1, \pm i \pm j, \pm k\}$	$\{\pm 1\}$

5. If H and K are normal subgroups of a group G then HK is also normal subgroup of G .
6. Let H be a subgroup of G . Then H is a normal subgroup of $N_G(H)$.
7. Centralizer $C_G(H)$ of a normal subgroup H of G is normal in G .
8. A subgroup H of a group G is normal in G if and only if every generator of G transforms each generator of H into an element of H .
9. Let H be a normal subgroup of G . Then $N_G(H) = G$.
10. A cyclic group G whose order is a prime number is simple.
11. The alternating group A_n of degree n is a normal subgroup of S_n and has order $\frac{n!}{2}$.
12. A_n is simple for all $n \geq 5$.

Some Important Theorems Of Group Theory

Lagrange's Theorem:

The order and index of a subgroup of a finite group divide the order of that group.

First Isomorphism Theorem:

Let $\phi : G \longrightarrow G'$ be an epimorphism.

$$\text{Then } \frac{G}{\text{Ker } \phi} \cong G'$$

Second Isomorphism Theorem:

Let H be a normal subgroup of G and K be a subgroup of a group G . Then

$$\frac{HK}{H} \cong \frac{K}{H \cap K}$$

Third Isomorphism Theorem:

Let H, K be normal subgroups of G and $H \subseteq K$. Then

$$\frac{\frac{G}{H}}{\frac{K}{H}} \cong \frac{G}{K}$$

Cauchy's Theorem for Abelian group:

If G is a finite Abelian group p a prime divisor of the order of G then G contains an element of order p .

Cauchy's Theorem for Non-Abelian group:

If a prime p divides the order of a group G then G contains an element of order p .

Sylow's First Theorem:

A finite group whose order is divisible by a prime p contains a Sylow p -subgroup.

Sylow's Second theorem:

Any two sylow p-subgroups of a group are conjugate.

Sylow's Third Theorem:

The number k of sylow p-subgroups of a finite group is congruent to 1 mod p and is a factor of the order of the group.

Table of Isomorphic Groups:

The following table shows the information of number of isomorphic groups of different orders.

Order	Abelian	Non-Abelian	Total
1	$\langle e \rangle$	-	1
2	C_2	-	1
3	C_3	-	1
4	$C_4, C_2 \times C_2$	-	2
5	C_5	-	1
6	C_6	D_3	2
7	C_7	-	1
8	$C_8, C_2 \times C_4, C_2 \times C_2 \times C_2$	D_4, Q_8	5
9	$C_9, C_3 \times C_3$	-	2
10	C_{10}	D_5	2
11	C_{11}	-	1
12	$C_{12}, C_2 \times C_6$	A_4, D_6, T	5
13	C_{13}	-	1
14	C_{14}	D_7	2
15	C_{15}	-	1
16	$C_{16}, C_8 \times C_2, C_4 \times C_4, C_4 \times C_2 \times C_2, C_2 \times C_2 \times C_2 \times C_2$	$D_8, D_4 \times C_2, Q \times C_2, G_{16}^{(4)}, G_{16}^{(5)}, G_{16}^{(6)}, G_{16}^{(7)}, G_{16}^{(8)}, G_{16}^{(9)}$	14
17	C_{17}	-	1
18	$C_{18}, C_6 \times C_3$	$D_9, S_3 \times C_3, G_{18}^{(3)}$	5
19	C_{19}	-	1

20	$C_{20}, C_{10} \times C_2$	$D_{10}, F_{20}, G_{20}^{(2)}$	5
21	C_{21}	F_{21}	2
22	C_{22}	D_{11}	2
23	C_{23}	-	1
24	$C_{24}, C_2 \times C_{12},$ $C_2 \times C_2 \times C_6$	$S_4, S_3 \times C_4, S_3 \times C_2 \times C_2,$ $D_4 \times C_3, Q \times C_3, A_4 \times C_2,$ $T \times C_2$, plus 5 others	15
25	$C_{25}, C_5 \times C_5$	-	2
26	C_{26}	D_{13}	2
27	$C_{27}, C_9 \times C_3, C_3 \times C_3 \times C_3$	$G_{27}^{(1)}, G_{27}^{(2)}$	5
28	$C_{28}, C_2 \times C_{14}$	$D_{14}, G_{28}^{(2)}$	4
29	C_{29}	-	1
30	C_{30}	$D_{15}, D_5 \times C_3, D_3 \times C_5$	4
31	C_{31}	-	1

Multiple Choice Questions

1. If X and Y are two sets such that $n(X) = 17$, $n(Y) = 23$ and $n(X \cup Y) = 38$, then $n(X \cap Y)$ is
(a) 4 (b) 2 (c) 6 (d) 8
2. Let U be the universal set and A, B are subsets of U . If $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then $n(A' \cap B') =$
(a) 400 (b) 240 (c) 300 (d) 600
3. Which of the following is the empty set
(a) $\{x \mid x \in R \wedge x^2 - 1 = 0\}$ (b) $\{x \mid x \in R \wedge x^2 + 1 = 0\}$
(c) $\{x \mid x \in R \wedge x^2 - 16 = 0\}$ (d) $\{x \mid x \in R \wedge x^2 = x + 2\}$
4. If X and Y are two sets, then $X \cap (X \cup Y)' =$
(a) X (b) Y (c) ϕ (d) $X \cap Y$
5. If a $N = \{ax \mid x \in N\}$, then $3N \cap 5N =$
(a) N (b) $3N$ (c) $15N$ (d) $12N$
6. Let r be a relation defined on a set of real number as $\forall a, b \in R, a r b$ iff $|a - b| > 0$. Then r is
(a) Reflexive (b) symmetric (c) transitive (d) None
7. n/m means that n is factor of m , then the relation ' $/$ ' is
(a) symmetric (b) reflexive
(c) transitive and reflexive (d) equivalence
8. Let r_1 and r_2 are symmetric relations (not disjoint) on a set A then the relation $r_1 \cap r_2$ is
(a) reflexive (b) symmetric (c) transitive (d) Anti-symmetric
9. Let r be a relation defined as $a r b$ if $|a| \leq b$. Then relation r is
(a) reflexive (b) symmetric
(c) transitive (d) reflexive and symmetric
10. Let r is be a relation defined as $a r b$ iff $ab + 1 > 0$. Then relation r is
(a) reflexive (b) symmetric
(c) transitive (d) reflexive and symmetric
11. Number of non-empty subsets of the set $\{1, 2, 3, 4\}$ is
(a) 15 (b) 16 (c) 17 (d) 14
12. Let $n(A) = 3$ and $n(B) = 6$. What can be the minimum number of elements in $A \cup B$?
(a) 18 (b) 9 (c) 6 (d) 3
13. The relation 'less than' in the set of natural number is
(a) only symmetric (b) only transitive
(c) only reflexive (d) None
14. The union of all positive even and all positive odd integer is
(a) N (b) W (c) Z (d) Z^+

15. Which binary operation is not defined in the set of natural number
(a) addition (b) subtraction (c) multiplication (d) None
16. Which of the following is not a group w.r.t. addition
(a) N (b) Z (c) Q (d) R
17. Which of the following is a group w. r. t multiplication
(a) Z (b) Q (c) $Q - \{0\}$ (d) R
18. The identity element of $(P(X), \cap)$ where $X \neq \phi$ is
(a) X (b) ϕ (c) $P(X)$ (d) None
19. $G = \{1, 3, 5, 7\}$ is a group w. r. t multiplication modulo 8. Then inverse of 5 is
(a) $\frac{1}{5}$ (b) 5 (c) 7 (d) 3
20. $G = \{1, 2, 4, 5, 7, 8\}$ under multiplication modulo 9 is
(a) semi-group (b) monoid (c) group (d) Abelian
21. Let G be a group and $a, b \in G$. Then order of a^{-1} is equal to
(a) b (b) ab (c) bab^{-1} (d) b^{-1}
22. In a group of even order, there at least _____ elements of order 2.
(a) one (b) two (c) three (d) none
23. If G is an Abelian, then which of the following is the order of G .
(a) 5 (b) 6 (c) 7 (d) 24
24. The set of cube roots of unity is a subgroup of
(a) R (b) C (c) $C - \{0\}$ (d) R^+
25. The number of subgroups of a group is
(a) one (b) at least two (c) at most two (d) two
26. Let H and K be subgroups G . Then which of the following is a subgroups of G .
(a) $H + \{0\}$ (b) $H \cap K$ (c) $H - K$ (d) HUK
27. Let H and K be subgroup of G . Then HUK is also a subgroups of G . if
(a) $H \cap K = \phi$ (b) $H \cap K \neq \phi$ (c) $H \subset K$ (d) $H - K = \phi$
28. Which of the following is a subgroup of group $G = \{0, 1, 2, 3\}$ w.r.t. addition modulo 4.
(a) $\{0, 1\}$ (b) $\{0, 2\}$ (c) $\{1, 2\}$ (d) $\{0, 1, 3\}$
29. Let G be a cyclic group. Then subgroup H of G is
(a) Alelian (b) cyclic
(c) generated by single element (d) All of these
30. Let G be a cyclic group of order 24 and H be a subgroup of G . Then order of H is
(a) 5 (b) 7 (c) 8 (d) 10
31. Let G be a cyclic group of order 17. Then number of subgroups of G is
(a) 2 (b) 8 (c) 17 (d) 34
32. Let G be a cyclic group of order 10. Then number of subgroups of G is
(a) 2 (b) 5 (c) 4 (d) 10

33. The number of distinct left cosets of a subgroup H of a group G is called _____ of H in G .
(a) Index (b) order (c) element (d) partition
34. Let G be a finite group. Let H be a subgroup of G . Then which of the following divides order of G .
(a) order of H (b) index of H (c) order of G (d) All of these
35. Let G be a group of order 36 and let $a \in G$. The order of a is
(a) 11 (b) 15 (c) 18 (d) 21
36. Let G be a group of order 37 and $a \in G$. Then order of a is
(a) even number (b) odd number
(c) prime number (d) composite number
37. Let G be a group of order prime number. Then
(a) G is abelian (b) G is trivial
(c) G has 3 subgroups (d) None
38. Which of the following is abelian but not cyclic
(a) $\mathbb{Z}$ (b) $\{1, w, w^2\}$
(c) $\{\pm 1, \pm i\}$
(d) $G = \{1, 3, 5\}$ w. r. t multiplication modulo 6.
39. Let G be a group which have no proper subgroup. Then order of G is
(a) 15 (b) 14 (c) 47 (d) 81
40. Let G be group and H be subgroup of G of order 8. Then order of G is
(a) 22 (b) 32 (c) 42 (d) 52
41. Let G be infinite cyclic group. Then number of generator of G is
(a) 1 (b) 2 (c) 3 (d) 4
42. Which of the following is cyclic group.
(a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) $\mathbb{C}$
43. Let G be a cyclic group of order 24. Then order of a^4 is
(a) 2 (b) 4 (c) 6 (d) 8
44. Let G be a cyclic group of order 24. Then order of a^9 is
(a) 2 (b) 4 (c) 6 (d) 8
45. Let H, K be two subgroup of a group G . Then set $HK = \{hk \mid h \in H \wedge k \in K\}$ is a subgroup of G if
(a) G is Abelian (b) G is cyclic
(c) G is of prime number (d) All of these
46. let H, K be two subgroup of a finite group G . Then for any $g \in G$ which of the following is true.
(a) $gH = Kg$ (b) $g(H \cap K) = gH \cap gK$
(c) $g(H \cap K) = (H \cap K)g$ (d) $g(H \cup K) = gH \cup gK$
47. Let $G = \{1, 2, 3, \dots, 12\}$ be a group w. r. t multiplication modulo 13. Then which of the following is a subgroup of G .
(a) $H = \{1, 3, 5, 7, 9, 11\}$ (b) $K = \{1, 2, 4, 6, 8, 10, 12\}$
(c) $M = \{1, 6, 8, 10\}$ (d) $T = \{1, 3, 9\}$

48. Let X be a non-empty set. The permutation on X is a
(a) bijective mapping (b) injective mapping
(c) surjective mapping (d) inverses mapping
49. Let X has n elements. Then number of bijective mapping on X is
(a) n^2 (b) $(n)^n$ (c) $n!$ (d) $n \times n$
50. Let X has n elements. Then number of mapping on X is
(a) n^2 (b) $(n)^n$ (c) $n!$ (d) $n \times n$
51. Let X has n elements. The set S_n of all permutations on X is a group w. r. to mappings
(a) additions (b) multiplication
(c) compositions (d) Inverse
52. The group S_n is called
(a) symmetric group of degree n (b) dihedral group of degree n
(c) polynomial group of degree n (d) None
53. Let $X = \{1, 2, 3\}$. Then S_3 has _____ elements
(a) 3 (b) 4 (c) 6 (d) 9
54. Which of the following is Abelian
(a) S_2 (b) S_3 (c) S_4 (d) S_5
55. A group G is Abelian. Then
(a) $|G| \leq 9$ (b) $|G| \leq 5$ (c) $|G| \leq 3$ (d) $|G| \leq 12$
56. Which of the following is a representation of S_3
(a) $\{ \langle a, b \rangle \mid a^3 = b^2 = (ab)^2 = e \}$ (b) $\{ \langle a, b \rangle \mid a^2 = b^2 = e \}$
(c) $\{ \langle a \rangle \mid a^2 = e \}$ (d) $\{ \langle a, b \rangle \mid a^2 = b^2 = (ab)^2 = e \}$
57. Which of the following is a cyclic permutation
(a) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$
(c) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$ (d) None
58. Product of two cyclic permutations is
(a) Even permutation (b) odd permutation
(c) cyclic permutation (d) not a cyclic permutation
59. A transposition is a cycle of length
(a) 1 (b) 2 (c) 3 (d) 4
60. A permutation of degree n can be expressed as a product of
(a) cycles (b) transposition
(c) permutations (d) All of these
61. Which of the following is even permutation.
(a) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$
(c) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$ (d) None

62. In S_4 group of permutation, number of even permutation is
 (a) 4 (b) 12 (c) 16 (d) 24
63. In S_n number of odd permutation is
 (a) $n!$ (b) $\frac{n!}{3}$ (c) $\frac{n!}{2}$ (d) $\frac{n+1}{2}$
64. The A_n , set of all even permutations of S_n is a subgroup of S_n . Then order of A_n is
 (a) $n!$ (b) $\frac{n!}{3}$ (c) $\frac{n!}{2}$ (d) $\frac{n+1}{2}$
65. In S_3 $a = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$. Then a^{-1} is
 (a) $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$
 (c) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ (d) None
66. Which of the following is a subgroup of $S_3 = \{a, a^2, b, ab, a^2b, e\}$.
 (a) $\{e, a, a^2\}$ (b) $\{e, b\}$ (c) $\{e, a^2, b\}$ (d) All of These
67. The order of permutation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$ is
 (a) 1 (b) 2 (c) 3 (d) 4
68. The order of permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 1 & 6 & 5 \end{pmatrix}$ is
 (a) 1 (b) 2 (c) 3 (d) 4
69. The only idempotent element in a group is
 (a) inverse element (b) identity
 (c) element of order of two (d) finite order element
70. A group all of whose elements are of finite order is called
 (a) periodic group (b) symmetric group
 (c) finite group (d) local group
71. A group in which every element except the identity has infinite order is called
 (a) torsion free (b) a-periodic
 (c) locally infinite (d) all of these
72. If a group is neither periodic nor torsion free, then G is
 (a) mixed group (b) infinite group
 (c) free group (d) symmetric group
73. Which of the following group with binary operation of ordinary addition is torsion free.
 (a) $\mathbb{Z}$ (b) $\mathbb{R}$ (c) $\mathbb{C}$ (d) all of these
74. Let $D_4 = \{ \langle a, b \rangle; a^4 = b^2 = (ab)^2 = 1 \}$ be a dihedral group of order 8. Then which of the following is a subgroup of D_4 .
 (a) $\{ \langle a, b \rangle; (ab)^2 = 1 \}$ (b) $\{ \langle a, b \rangle; a^4 = b^2 = 1 \}$
 (c) $\{ \langle a^3, b \rangle; (a^3b)^2 = 1 \}$ (d) $\{ \langle a, b \rangle; b^2 = 1 \}$

75. A mapping $\phi: G \rightarrow G'$ is called homomorphism if for $a, b \in G$
 (a) $\phi(ab) = \phi(a) + \phi(b)$ (b) $\phi(ab) = \phi(a)\phi(b)$
 (c) $\phi(ab) = \phi(a) - \phi(b)$ (d) $\phi(ab) = \phi(a)\phi(b)^{-1}$
76. If a homomorphism is also subjective then it is called
 (a) monomorphism (b) epimorphism
 (c) isomorphism (d) None
77. A monomorphism is a homomorphism which is also
 (a) injective (b) subjective (c) bijective (d) onto
78. A bijective homomorphism is called
 (a) epimorphism (b) monomorphism
 (c) isomorphism (d) None
79. The relation of isomorphism between group is
 (a) symmetric relation (b) transitive relation
 (c) equivalence relation (d) total order relation
80. Let E be a group of even integers under binary operation of addition. Then which of the following is isomorphic to E
 (a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) $\mathbb{C}$
81. $\mathbb{R}^+$ is a group of non-zero positive real number under multiplication. Then which of the following group under addition is isomorphic to $\mathbb{R}^+$.
 (a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) $\mathbb{C}$
82. $\phi: \mathbb{R}^+ \longrightarrow \mathbb{R}$ is an isomorphism. Then $\forall x \in \mathbb{R}^+$ which of the following is true.
 (a) $\phi(x) = x$ (b) $\phi(x) = x^2 + 1$
 (c) $\phi(x) = \log(x)$ (d) $\phi(x) = \tan(x)$
83. The homomorphic image $\phi(G)$ of a group G is
 (a) groupoid (b) monoid
 (c) group (d) Abelian
84. Any group G can be embedded in a group of bijective mappings of certain set is a statement of
 (a) Lagrange's theorem (b) Cauchy's theorem
 (c) Cayley's theorem (d) Isomorphism theorem
85. Which of the following is a representation of $C_4 = \{1, -1, i, -i\}$
 (a) $\langle x: x^4 = 1 \rangle$ (b) $\langle a, b: a^2 = b^2 = (ab)^2 = 1 \rangle$
 (c) $\langle a, b: a^3 = b^2 = (ab)^2 = 1 \rangle$ (d) $\langle a, b: a^4: b^2 = 1 \rangle$
86. Let $G = \langle a, b; a^{-1}b^2a = b^3, b^{-1}a^2b = a^3 \rangle$ Then G is
 (a) S_3 (b) D_8
 (c) Identity group (d) C_4
87. Which of the following is a Klein's four group
 (a) $\langle a, b; (ab)^2 = 1 \rangle$ (b) $\langle a, b; a^2 = b^2 = 1 \rangle$
 (c) $\langle a^3, b; (a^3b)^2 = 1 \rangle$ (d) $\langle a, b; a^4 = b^2 (ab)^2 = 1 \rangle$

88. Let $G = \langle a, b : a^3 = b^2 = (ab)^2 = 1 \rangle$ Then G is
 (a) V_4 (b) S_3 (c) D_8 (d) S_4
89. Which of the following is a representation of group of quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$.
 (a) $\langle a, b; a^4 = (ab)^2 = b^2 = 1 \rangle$ (b) $\langle a, b; a^4 = 1, a^2 = b^2 = (ab)^2 = e \rangle$
 (c) $\langle a, b; a^4 = b^2 = (ab)^4 \rangle$ (d) None
90. Let G & H be two cyclic group of order m and n respectively. If G & H are isomorphic then
 (a) $m < n$ (b) $m > n$
 (c) $m = n$ (d) $m \neq n$
91. Let G be a cyclic group. Then which of the following can be order of G .
 (a) 17 (b) 107
 (c) 1,000,000,007 (d) all of these
92. Let G be a cyclic group. Then which of the following is also cyclic
 (a) subgroup of G (b) homomorphic image of G
 (c) centre of G (d) all of these
93. For any set of points S in a plane, the set of all distance preserving injective mappings of a plane which leave the points of S invariant is called
 (a) mappings (b) Invariant mappings
 (c) symmetries (d) Permutations
94. The symmetries of a rectangle form a
 (a) Kleins four group, V_4 (b) Dihedral group of order 8, D_4
 (c) Optic group (d) Permutation group of order 3, S_3
95. The symmetries of square of form a
 (a) Klein's four group, V_4 (b) Dihedral group of order 8, D_4 (Optic group)
 (c) Group of quaternian (d) Permutation group of order 3, S_3
96. The symmetries of an equilateral triangle form a
 (a) Klein's four group, V_4 (b) Dihedral group of order 8, D_4
 (c) Optic group (d) Permutation group of order 3, S_3
97. The symmetries of a n -polygon form a
 (a) Dihedral group of order $2n$, D_n (b) Permutation group of order n , S_n
 (c) cyclic group of order n , C_n (d) None
98. For dihedral group of order $2n$, D_n , if $n = 2$, $D_2 =$
 (a) Klein's four group (b) S_4
 (c) C_4 (d) None
99. For dihedral group of order $2n$, D_n , which of the following is true
 (a) $D_2 = V_4$ (b) $D_3 = S_3$ (c) (a) & (b) (d) None
100. The smallest non-cyclic group is
 (a) V_4 (b) S_3 (c) D_4 (d) None

101. The smallest non-abelian group is
 (a) V_4 (b) S_3 (c) D_4 (d) None
102. In $S_3 = \langle a, b \mid a^3 = b^2 = (ab)^2 = 1 \rangle$. The number of distinct right cosets of subgroup $\{e, a, a^2\}$ are
 (a) 1 (b) 2 (c) 3 (d) 4
103. In a dihedral group of order 8, $D_4 = \langle a, b; a^4 = b^2 = (ab)^2 = 1 \rangle$, the number of distinct right cosets of subgroup $H = \langle a: a^4 = 1 \rangle$ are
 (a) 1 (b) 2 (c) 3 (d) 4
104. Let H and K be two left (right) cosets of a subgroup of a group. Then which of the following is true
 (a) either $H = K$ or $H \cap K = \phi$ (b) $H \cap K \neq \phi$
 (c) either $H \subseteq K$ or $K \subseteq H$ (d) $H \cap K = \{e\}$
105. Let H and K be two subgroup of G . Let index of $H = n$ & index of $K = m$, then index of $(H \cap K) =$
 (a) n (b) m (c) nm (d) finite
106. Let X be an ordinary subset of a group G . Then the set $\{a \in G : aX = Xa\}$ is
 (a) Normalizer of X in G (b) Centralizer of X in G
 (c) Center of G (d) Commuter of X in G
107. Let $H = \{a, b\}$ in $S_3 = \{e, a, a^2, b, ab, a^2b\}$ Then $N_{S_3}(H) =$
 (a) H (b) $\{e, a, a^2\}$
 (c) $\{e, a^2b\}$ (d) G
108. Let $H = \{e, a, a^2\}$ in $S_3 = \{e, a, a^2, b, ab, a^2b\}$ Then $N_{S_3}(H) =$
 (a) H (b) $\{e, a, a^2, ab, b\}$
 (c) $\{e, a^2b\}$ (d) G
109. Let X be a subset of G . Then the set $\{a \in G: ax = xa \forall x \in X\}$ is called
 (a) Normalizer of X in G (b) Centralizer of X in G
 (c) Center of G (d) None
110. Let $X = \{1, a, a^2, a^3\}$ in $D_4 = \{1, a, a^2, a^3, b, ab, a^2b, a^3b\}$, Then $N_{D_4}(X) =$
 (a) X (b) D_4 (c) $\{1, a, a^2, b\}$ (d) $\{1, a, a^2, a^3, b\}$
111. Let $X = \{1, a, a^2, a^3\}$ in $D_4 = \{1, a, a^2, a^3, ab, a^2b, a^3b, b\}$ Then centralize of X in D_4 , $C_{D_4}(X) =$
 (a) X (b) D_4
 (c) $\{1, a^2\}$ (d) $\{1, a, a^2, a^3, b, a^2b\}$
112. Which of the following subset of dihedral group of order 8, $D_4 = \{1, a, a^2, a^3, ab, a^2b, a^3b, b\}$ has $N_{D_4}(X) = C_{D_4}(X)$
 (a) $X = \{1, a\}$ (b) $X = \{1, a^2\}$
 (c) $X = \{1, a, a^2, a^3\}$ (d) $X = \{1, a, b\}$

113. Let X be a subset of a group G such that $N_G(X) = C_G(X)$. Then $X =$
 (a) ϕ (b) singleton (c) finite (d) proper
114. Let X be subset of a group G . Then which of the following is a subgroup of G .
 (a) centralizer of X in G (b) Normalizer of X in G
 (c) center of G (d) all of these
115. Let X be a subset of a group G . If centralizer of X in $G =$ centre of G i.e. $C_G(X) = Z(G)$. Then which of the following is true
 (a) $X = \{e\}$ (b) $X = G$ (c) $G = \phi$ (d) None
116. Let G be an abelian group. Then
 (a) $Z(G) = \{e\}$ (b) $Z(G) = \phi$
 (c) $Z(G) = G$ (d) $Z(G) = X$
117. The group G is called a group without (trivial) center if
 (a) $Z(G) = \{e\}$ (b) $Z(G) = \phi$
 (c) $Z(G) = G$ (d) None
118. Let $Q = \{\pm 1, \pm i, \pm j, \pm k\}$ be a group of quaternion's. Then center of $Q = Z(Q) =$
 (a) $\{I\}$ (b) $\{\pm I\}$ (c) $\{I, i, j, k\}$ (d) Q
119. The center of S_3 is
 (a) empty (b) trivial (c) Non-trivial (d) None
120. Let G be a group such that $Z(G) = G$. Then $G =$
 (a) Z (b) Q (c) R (d) all of these
121. Let C be a group of complex number w. r. to addition. Then $Z(C) =$
 (a) Q (b) R (c) C (d) Z
122. Let $a, b \in G$ called conjugate elements if there exists $g \in G$ such that
 (a) $ag^{-1} = bg^{-1}$ (b) $agg^{-1} = b$
 (c) $ab = g$ (d) $a = gbg^{-1}$
123. The relation of conjugacy between elements of a group is
 (a) symmetric relation (b) total order relation
 (c) equivalence relation (d) lattice
124. Let H, K be two subgroup of G then H is subgroup conjugate to K if for each $g \in G$
 (a) $K = gHg^{-1}$ (b) $gKg^{-1} = H$
 (c) $Kg = gH$ (d) all of these
125. The relation of conjugacy between subgroups of a group is
 (a) symmetric relation (b) equivalence relation
 (c) total order relation (d) lattice
126. Let H and K be two conjugate subgroups of a group G . Then H and K are
 (a) finite (b) Isomorphic
 (c) infinite (d) symmetric
127. What is the order of group $G = \langle a, b; a^6 = b^2 = (ab)^2 = 1 \rangle$.
 (a) 6 (b) 10 (c) 12 (d) 24

128. Which of the following group is order 8.
- (a) $G = \langle a, b; a^4 = b^2 = (ab)^2 = 1 \rangle$ (b) $G = \langle a, b; a^2 = b^2 = (ab)^2, a^4 = 1 \rangle$
 (c) $G = \langle a, b; a^4 = b^2 = 1, ab = ba \rangle$ (d) all of these
129. Let H be a normal subgroup of G . The which of the following is true
- (a) $N_G(H) = H$ (b) $\forall g \in G, gHg^{-1} = H$
 (c) $\forall g \in G, gH = Hg$ (d) all of these
130. Let G be a group such that all of its subgroups are normal. Then G can be
- (a) Abelian group (b) Hamiltonion group
 (c) Identity group (d) all of these
131. The normal subgroup of $S_3 = \{e, a, a^2, b, ab, a^2b\}$ is
- (a) $\{e, b\}$ (b) $\{e, ab\}$ (c) $\{e, a^2b\}$ (d) $\{e, a, a^2\}$
132. Let H, K be two normal subgroup of G . Then which of the following is also a normal subgroup of G .
- (a) $H + K$ (b) $H - K$ (c) $H \cup K$ (d) $H \cap K$
133. Let H, K be normal subgroup of G . Then which of the following is also normal in G .
- (a) HK (b) $C_G(H)$ (c) $H \cap K$ (d) all of these
134. A subgroup of index 2 in a group is
- (a) normal (b) abelian (c) symmetric (d) none
135. Let H be a subgroup of G . then which of the following is a normal subgroup of $N_G(H)$
- (a) H (b) G (c) $\frac{G}{H}$ (d) None
136. Which of the following is a normal subgroup of Z w. r. to addition
- (a) $\langle n \rangle = \{kn: k \in Z\}$ (b) $\{1, -1\}$
 (c) N (d) W
137. Which of the following is a normal subgroup of Q of rational numbers w. r. to addition.
- (a) Z (b) N (c) W (d) R
138. Let $\phi: G \rightarrow G'$ is a homomorphism then the set $\{g \in G: \phi(g) = e', e' \in G'\}$ is
- (a) None-empty (b) Normal subgroup of G
 (c) $\text{Ker } \phi$ (d) all of these
139. In Fundamental theorem of homomorphism $\phi: G \rightarrow G'$ should be
- (a) mono-morphism (b) Isomorphism
 (c) bijective (d) epimorphism
140. If $\phi: G \rightarrow G'$ is an epimorphism then
- (a) $G \cong G'$ (b) $\frac{G}{\text{Ker } \phi} \cong G'$ (c) $\frac{G'}{\text{Ker } \phi} \cong G$ (d) None
141. Let $\phi: G \rightarrow G'$ be epimorphism such that $\text{ker } \phi = \{e\}$, then ϕ is
- (a) Isomorphism (b) identity mapping
 (c) ϕ is invertible mapping (d) None

142. Let H, K be subgroup of G such that H is normal in G . Then which of the following is true
 (a) HK is a subgroup of G (b) $H \cap K$ is normal in K
 (c) $\frac{HK}{H} \cong \frac{K}{H \cap K}$ (d) all of these
143. Let H, K be normal subgroup of G and $H \subseteq K$. Then $\frac{G/H}{K/H} \cong \frac{G}{K}$. This is a statement of
 (a) 1st isomorphism theorem (b) 2nd isomorphism theorem
 (c) 3rd isomorphism theorem (d) None
144. If G is an abelian group. The which of the following is true
 (a) centre of G , $Z(G)$ is normal in G (b) $\frac{G}{Z(G)}$ is cyclic in G
 (c) $Z(G) = G$ (d) all of these
145. Let G be an abelian group. Then which of the following can be order of G .
 (a) 5 (b) 25 (c) 49 (d) all of these
146. Let G be an abelian group. Then which of the following cannot be order of G .
 (a) 9 (b) 16 (c) 25 (d) 121
147. If $\phi : G \rightarrow G$ is homomorphism. Then ϕ is called
 (a) endomorphism (b) monomorphism
 (c) isomorphism (d) automorphism
148. Let $\phi: G \rightarrow G$ is a bijective homomorphism. Then ϕ is called
 (a) endomorphism (b) monomorphism
 (c) isomorphism (d) automorphism
149. let G be a group. The set of all automorphism $A(G)$ is a
 (a) semi group (b) group
 (c) normal subgroup (d) none
150. Let a be a fixed element of a group G . The mapping $I_a(g) = aga^{-1}$, $g \in G$ is
 (a) automorphism (b) Inner automorphism
 (c) outer automorphism (d) None
151. If all the automorphism of a group G except identity automorphism are outer automorphism, the G is
 (a) Abelian (b) normal (c) finite (d) None
152. Which of the following group has all of its automorphism are inner automorphism
 (a) S_3 (b) Z (c) Q (d) R
153. The set of all inner automorphism of a group G , $I(G)$ is normal subgroup of
 (a) G (b) $A(G)$ (c) $Z(G)$ (d) None
154. Let G be a group. The $\frac{G}{Z(G)} \cong$
 (a) G (b) $A(G)$ (c) $I(G)$ (d) $Z(G)$
155. In a group of quaternion $G = \{\pm 1, \pm i, \pm j, \pm k\}$ which of the following is true.
 (a) $I(G) \cong \frac{G}{Z(G)}$ (b) $\frac{G}{Z(G)} \cong C_2 \times C_2$

- (c) $I(G) \cong C_2 \times C_2$ (d) all of these
156. The only proper normal subgroup of S_4 is
(a) A_1 (b) A_2 (c) A_3 (d) A_4
157. A group G has no proper normal subgroup. Then G is called
(a) complete (b) improper
(c) simple (d) None
158. A_n is simple if
(a) $n \leq 4$ (b) $n \geq 5$
(c) $n \leq 5$ (d) $n \geq 12$
159. Which of the following is a sylow theorem
(a) A finite group whose order is divisible by a prime P contains a sylow P -subgroup
(b) Any two sylow P -subgroups are conjugate
(c) A finite group G has unique sylow P -subgroup H iff H is normal in G .
(d) All of these
160. If group G is not simple then order of G is
(a) 35 (b) 410 (c) 2540 (d) all of these

Key To MCQ Of Group Theory

1	b	2	c	3	b	4	c	5	C	6	b
7	c	8	b	9	c	10	d	11	A	12	c
13	b	14	a	15	b	16	a	17	C	18	b
19	b	20	d	21	c	22	a	23	A	24	c
25	b	26	b	27	c	28	b	29	D	30	c
31	a	32	c	33	a	34	d	35	C	36	c
37	a	38	d	39	c	40	b	41	B	42	a
43	c	44	d	45	d	46	b	47	D	48	a
49	c	50	b	51	c	52	a	53	C	54	a
55	b	56	a	57	a	58	d	59	B	60	d
61	b	62	b	63	c	64	c	65	C	66	d
67	d	68	b	69	b	70	a	71	D	72	a
73	d	74	c	75	b	76	b	77	A	78	c
79	c	80	a	81	c	82	c	83	C	84	c
85	a	86	c	87	b	88	b	89	B	90	c
91	c	92	d	93	c	94	a	95	C	96	d
97	a	98	a	99	c	100	a	101	B	102	b
103	d	104	a	105	d	106	a	107	A	108	d
109	b	110	a	111	b	112	b	113	B	114	d
115	b	116	c	117	a	118	b	119	B	120	d
121	c	122	d	123	c	124	d	125	B	126	b
127	c	128	d	129	d	130	d	131	D	132	d
133	d	134	a	135	a	136	a	137	A	138	d
139	d	140	b	141	a	142	d	143	C	144	d
145	d	146	b	147	a	148	d	149	B	150	b
151	a	152	a	153	b	154	c	155	D	156	d
157	c	158	c	159	d	160	d				